

Step 1: Install requests Library

Open VS Code terminal and run:

```
``bash id="z6q9jr" pip install requests
```

```
---  
  
# Step 2: views.py  
  
Open:  
  
```text id="0r0d0h"  
api/views.py
```

Add:

```
``python id="ykvq0m" import requests
```

```
from rest_framework.views import APIView from rest_framework.response import Response from
rest_framework import status
```

---

## 1. Music Weather API

---

```
class MusicWeatherAPIView(APIView):
```

```
def get(self, request, city):

 API_KEY = "YOUR_OPENWEATHER_API_KEY"

 url = (
 f"https://api.openweathermap.org/data/2.5/weather?"
 f"q={city}&appid={API_KEY}&units=metric"
)

 response = requests.get(url)
```

```

data = response.json()

if response.status_code != 200:

 return Response(
 {"error": "City not found"},
 status=status.HTTP_404_NOT_FOUND
)

result = {
 "city": city,
 "temperature": data["main"]["temp"],
 "description": data["weather"][0]["description"]
}

return Response(result)

```

---

## 2. Food Location API

---

```
class FoodLocationAPIView(APIView):
```

```

def get(self, request):

 restaurant = request.GET.get("restaurant")

 API_KEY = "YOUR_GOOGLE_MAPS_API_KEY"

 url = (
 "https://maps.googleapis.com/maps/api/geocode/json"
 f"?address={restaurant}&key={API_KEY}"
)

 response = requests.get(url)

 data = response.json()

 if not data["results"]:

```

```

 return Response(
 {"error": "Restaurant not found"},
 status=status.HTTP_404_NOT_FOUND
)

 location = data["results"][0]["geometry"]["location"]

 result = {
 "restaurant": restaurant,
 "latitude": location["lat"],
 "longitude": location["lng"]
 }

 return Response(result)

```

### 3. Country Info API

```
class CountryInfoAPIView(APIView):
```

```

 def get(self, request, country):

 url = f"https://restcountries.com/v3.1/name/{country}"

 response = requests.get(url)

 data = response.json()

 if response.status_code != 200:

 return Response(
 {"error": "Country not found"},
 status=status.HTTP_404_NOT_FOUND
)

 country_data = data[0]

 result = {
 "country": country,
 "capital": country_data["capital"][0],

```

```
 "population": country_data["population"]
 }

 return Response(result)
```

---

## 4. GitHub Repositories API

---

```
class GithubReposAPIView(APIView):
```

```
 def get(self, request, username):

 url = f"https://api.github.com/users/{username}/repos"

 response = requests.get(url)

 data = response.json()

 if response.status_code != 200:

 return Response(
 {"error": "GitHub user not found"},
 status=status.HTTP_404_NOT_FOUND
)

 repos = []

 for repo in data:

 repos.append(repo["name"])

 return Response({
 "username": username,
 "repositories": repos
 })
```

```
...
```

---

## Step 3: urls.py

Open: `text id="6lv1ae"`  
`api/urls.py`

Add:

```
python id="ltv1t9" from django.urls import path
```

```
from .views import (MusicWeatherAPIView, FoodLocationAPIView, CountryInfoAPIView,
GithubReposAPIView)
```

```
urlpatterns = [
```

```
 path(
 'music-weather/<str:city>',
 MusicWeatherAPIView.as_view()
),

 path(
 'food-location/',
 FoodLocationAPIView.as_view()
),

 path(
 'country-info/<str:country>',
 CountryInfoAPIView.as_view()
),

 path(
 'github-repos/<str:username>',
 GithubReposAPIView.as_view()
),

```

```
]``
```

---

## Step 4: OpenWeatherMap API Key

Create free account: [url](https://openweathermap.org/api) OpenWeatherMap <https://openweathermap.org/api> Copy API key and  
replace: `python id="r6mvx8"`  
`YOUR_OPENWEATHER_API_KEY`

---

## Step 5: Google Maps API Key

Open:

 Google Cloud Console  <https://console.cloud.google.com/> 

Enable:

- Geocoding API

Copy API key and replace:

```
python id="vrz3v5" YOUR_GOOGLE_MAPS_API_KEY
```

```

Step 6: Run Server

```bash id="mnrr6v"  
python manage.py runserver
```

Step 7: Test APIs

Music Weather API

```
text id="5q2dru" http://127.0.0.1:8000/api/music-weather/London/
```

Example Response:

```
```json id="e08mji"  
{
 "city": "London",
 "temperature": 24,
 "description": "clear sky"
}
```

---

### Food Location API

```
text id="cz5eq6" http://127.0.0.1:8000/api/food-location/?restaurant=Dominos
```

Example Response:

```
```json id="ip8n0n"
{
  "restaurant": "Dominos",
  "latitude": 22.3039,
  "longitude": 70.8022
}
```

Country Info API

```text id="yv8trv" http://127.0.0.1:8000/api/country-info/India/

Example Response:

```
```json id="ubvh4z"
{
  "country": "India",
  "capital": "New Delhi",
  "population": 1400000000
}
```

GitHub Repositories API

```text id="afm2eu" http://127.0.0.1:8000/api/github-repos/octocat/

Example Response:

```
```json id="tw6l6k"
{
  "username": "octocat",
  "repositories": [
    "Hello-World",
    "Spoon-Knife"
  ]
}
```

Step 8: Postman AI Test

Open Postman.

Test:

```
```text id="2z6q3p" http://127.0.0.1:8000/api/music-weather/London/
```

Go to:

```
```text id="x17kdf"
Tests
```

Paste:

```
```javascript id="u5v2sw" pm.test("Check temperature exists", function () { var jsonData =
pm.response.json(); pm.expect(jsonData).to.have.property("temperature"); });
```

```
pm.test("Check description exists", function () { var jsonData = pm.response.json();
pm.expect(jsonData).to.have.property("description"); });
```

Click:

```
```text id="nchxq9"
Send
```

If both tests pass:

- Green success message appears.